CSEC Mathematics

January 2017 – Paper 2

Solutions

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SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, calculate the EXACT value of:

(i) $\dfrac{3\frac{1}{2} \times 1\frac{2}{3}}{4\frac{1}{5}}$ [2]

$$\text{Numerator} = 3\frac{1}{2} \times 1\frac{2}{3}$$

$$= \frac{7}{2} \times \frac{5}{3}$$

$$= \frac{35}{6}$$

$$\text{Denominator} = 4\frac{1}{5}$$

$$= \frac{21}{5}$$

$$\text{Numerator} \div \text{Denominator} = \frac{35}{6} \div \frac{21}{5}$$

$$= \frac{35}{6} \times \frac{5}{21}$$

$$= \frac{25}{18}$$

$$\therefore \frac{3\frac{1}{2} \times 1\frac{2}{3}}{4\frac{1}{5}} = \frac{25}{18}$$

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(ii) $5.47 - \sqrt{\frac{0.1014}{1.5}}$ [2]

Using the calculator,

$5.47 - \sqrt{\frac{0.1014}{1.5}} = 5.21$ (in exact form)

(b) The table below shows the number of tickets sold for a bus tour. Some items in the table are missing.

Tickets Sold for Bus Tour			
Category	**Number of Tickets Sold**	**Cost per Ticket in $**	**Total Cost in $**
Juvenile	5	P	130.50
Youth	14	44.35	Q
Adult	R		2483.60

(i) Calculate the value of P. [1]

5 Juvenile tickets cost = \$130.50

1 Juvenile ticket cost $= \frac{\$130.50}{5}$

$= \$26.10$

$\therefore P = 26.10$

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(ii) Calculate the value of Q. [1]

1 Youth ticket cost = \$44.35

14 Youth tickets cost = \$44.35 × 14

= \$620.90

$\therefore Q = 620.90$

(iii) An adult ticket is TWICE the cost of a youth ticket.

Calculate the value of R. [2]

An adult ticket costs twice as much as the cost of a Youth ticket.

Hence, the cost of an adult ticket = \$44.25 × 2

= \$88.70

The total cost of R adult tickets is \$2483.60.

Number of adult tickets sold, $R = \frac{\$2483.60}{\$88.70}$

$= 28$

$\therefore R = 28$

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(iv) The bus company pays taxes of 15% on each ticket sold.

Calculate the taxes paid by the bus company. [3]

Total cost = \$130.50 + \$620.90 + \$2483.60

= \$3235.00

The bus company pays taxes of 15% on each ticket sold.

Taxes paid = 15% of \$3235

$= \frac{15}{100} \times \$3235$

= \$485.25

∴ The bus company paid \$485.25 in taxes.

Total: 11 marks

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2. (a) Write as a single fraction:

$$\frac{2x+3}{3} + \frac{x-4}{4}$$ [2]

$$\frac{2x+3}{3} + \frac{x-4}{4}$$

$$= \frac{4(2x+3) + 3(x-4)}{12}$$

$$= \frac{8x + 12 + 3x - 12}{12}$$

$$= \frac{11x}{12}$$

(b) Write the following statement as an algebraic expression.

The sum of a number and its multiplicative inverse is five times the number. [2]

Statement: The sum of a number and its multiplicative inverse is five times the number.

Let the number be x.

Its multiplicative inverse is $\frac{1}{x}$.

Expression: $x + \frac{1}{x} = 5x$

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(c) Factorize completely:

(i) $x^2 - 36$ [2]

$x^2 - 36$

$= (x)^2 - (6)^2$ [difference of two squares]

$= (x - 6)(x + 6)$

(ii) $2x^2 + 5x - 12$ [2]

$2x^2 + 5x - 12$

$= 2x^2 + 8x - 3x - 12$

$= 2x(x + 4) - 3(x + 4)$

$= (2x - 3)(x + 4)$

(d) The formula for the volume of a cylinder is given as $V = \pi r^2 h$.

Make r the subject of the formula. [2]

$V = \pi r^2 h$

$\pi r^2 h = V$

$r^2 = \frac{V}{\pi h}$

$r = \sqrt{\frac{V}{\pi h}}$

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(e) Given that $x^2 + ax + b = (x+2)^2 - 3$, work out the values of a and b. [2]

$$(x+2)^2 - 3 = (x+2)(x+2) - 3$$

$$= x^2 + 4x + 4 - 3$$

$$= x^2 + 4x + 1$$

So, we have,

$$x^2 + ax + b = x^2 + 4x + 1$$

Comparing and equating coefficients gives:

$a = 4$ and $b = 1$

Total: 12 marks

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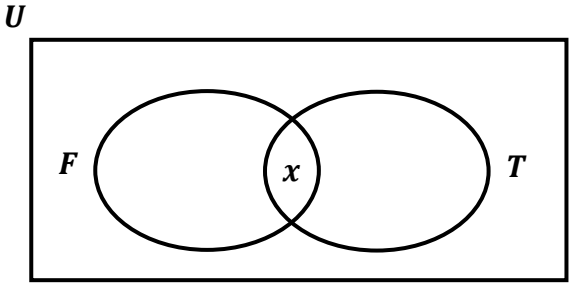
3. (a) The incomplete Venn diagram below shows the number of students in a class of 28 who play football and tennis.



$\boldsymbol{U}$ = {all students in the class}

$\boldsymbol{F}$ = {students who play football}

$\boldsymbol{T}$ = {students who play tennis}

Additional information about the class is that

12 students play tennis

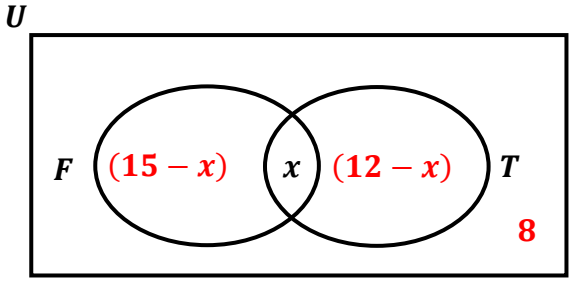
15 students play football

8 students play neither football nor tennis

x students play BOTH football and tennis.

(i) Complete the Venn diagram above to represent the information, showing the number of students in EACH subset. [3]

The complete Venn diagram is shown below:



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(ii) Calculate the value of x. [2]

The number of students in the class is 28.

Hence,

$$(15 - x) + x + (12 - x) + 8 = 28$$

$$35 - x = 28$$

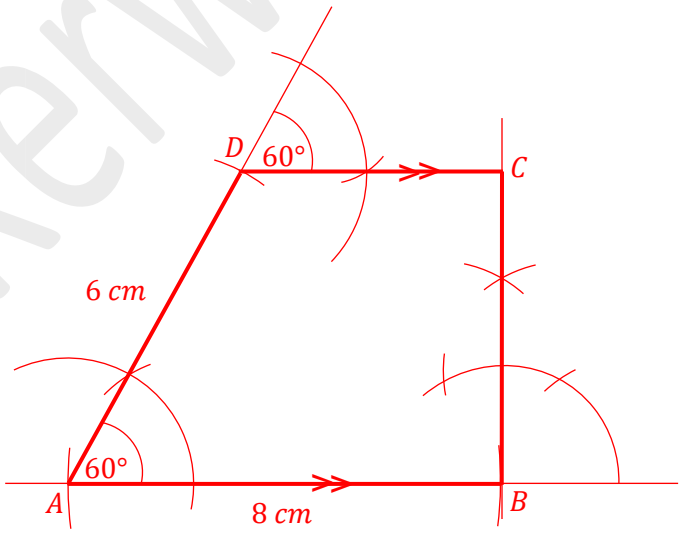
$$x = 35 - 28$$

$$x = 7$$

(b) Using **a ruler, a pencil and a pair of compasses**, construct the trapezium $ABCD$ with $AB = 8$ cm, $B\hat{A}D = 60°$, $AD = 6$ cm, $A\hat{B}C = 90°$ and AB parallel to CD.

(Credit will be given for clearly drawn construction lines.) [7]

The construction of the trapezium $ABCD$ is given below



Total: 12 marks

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4. (a) Given that $f(x) = 4x - 7$ and $g(x) = \frac{3x+1}{2}$, determine the values of:

(i) $g(0) + g(5)$ [2]

$g(x) = \frac{3x+1}{2}$

Now,

$g(0) + g(5) = \frac{3(0)+1}{2} + \frac{3(5)+1}{2}$

$= \frac{0+1}{2} + \frac{15+1}{2}$

$= \frac{1}{2} + \frac{16}{2}$

$= \frac{1}{2} + 8$

$= 8\frac{1}{2}$

(ii) $fg(5)$ [2]

$g(x) = \frac{3x+1}{2}$

$g(5) = \frac{3(5)+1}{2}$

$= \frac{15+1}{2}$

$= \frac{16}{2}$

$= 8$

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$f(x) = 4x - 7$

Hence,

$fg(5) = f[g(5)]$

$= f(8)$

$= 4(8) - 7$

$= 32 - 7$

$= 25$

$\therefore fg(5) = 25$

(iii) $f^{-1}(1)$ [2]

$f(x) = 4x - 7$

Let $y = f(x)$.

$y = 4x - 7$

Interchanging variables x and y:

$x = 4y - 7$

Making y the subject of the formula:

$x + 7 = 4y$

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$\frac{x+7}{4} = y$

Hence, $f^{-1}(x) = \frac{x+7}{4}$

Now,

$f^{-1}(1) = \frac{1+7}{4}$

$= \frac{8}{4}$

$= 2$

$\therefore f^{-1}(1) = 2$

(b) $P(6, -1)$ and $Q(2, 7)$ are the end point of a line segment PQ. Determine

(i) the gradient of PQ [2]

Points are $P(6, -1)$ and $Q(2, 7)$

Gradient of $PQ = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{7-(-1)}{2-6}$

$= \frac{8}{-4}$

$= -2$

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(ii) the coordinates of the midpoint of PQ [2]

Points are $P(6, -1)$ and $Q(2, 7)$

Midpoint of $PQ = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$= \left(\frac{6+2}{2}, \frac{-1+7}{2}\right)$

$= \left(\frac{8}{2}, \frac{6}{2}\right)$

$= (4, 3)$

(iii) the equation of the perpendicular bisector of PQ. [2]

Gradient of $PQ = -2$

Gradient of perpendicular bisector of $PQ = \frac{-1}{-2}$

$= \frac{1}{2}$

Substituting $m = \frac{1}{2}$ and point $(4, 3)$ into $y - y_1 = m(x - x_1)$ gives:

$y - 3 = \frac{1}{2}(x - 4)$

$y - 3 = \frac{1}{2}x - 2$

$y = \frac{1}{2}x - 2 + 3$

$y = \frac{1}{2}x + 1$

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$\therefore$ The equation of the perpendicular bisector of PQ is: $y = \frac{1}{2}x + 1$

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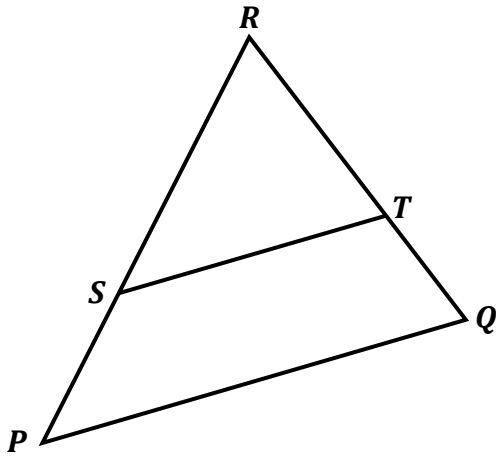
5. (a)



Triangles PQR and STR are similar triangles.

(i) Complete the following statement:

In the diagram above, the corresponding angles of ΔPQR and ΔSTR are equal and the ratio of their corresponding sides are the same. [2]

In the diagram above, not drawn to scale, $RS = 15\ cm, SP = 9\ cm$ and $ST = 12\ cm$.

(ii) Determine the length of PQ. [3]

$$\frac{PQ}{ST} = \frac{RP}{RS}$$

$$\frac{PQ}{12} = \frac{24}{15}$$

$$PQ = \frac{24 \times 12}{15}$$

$$PQ = 19.2\ cm$$

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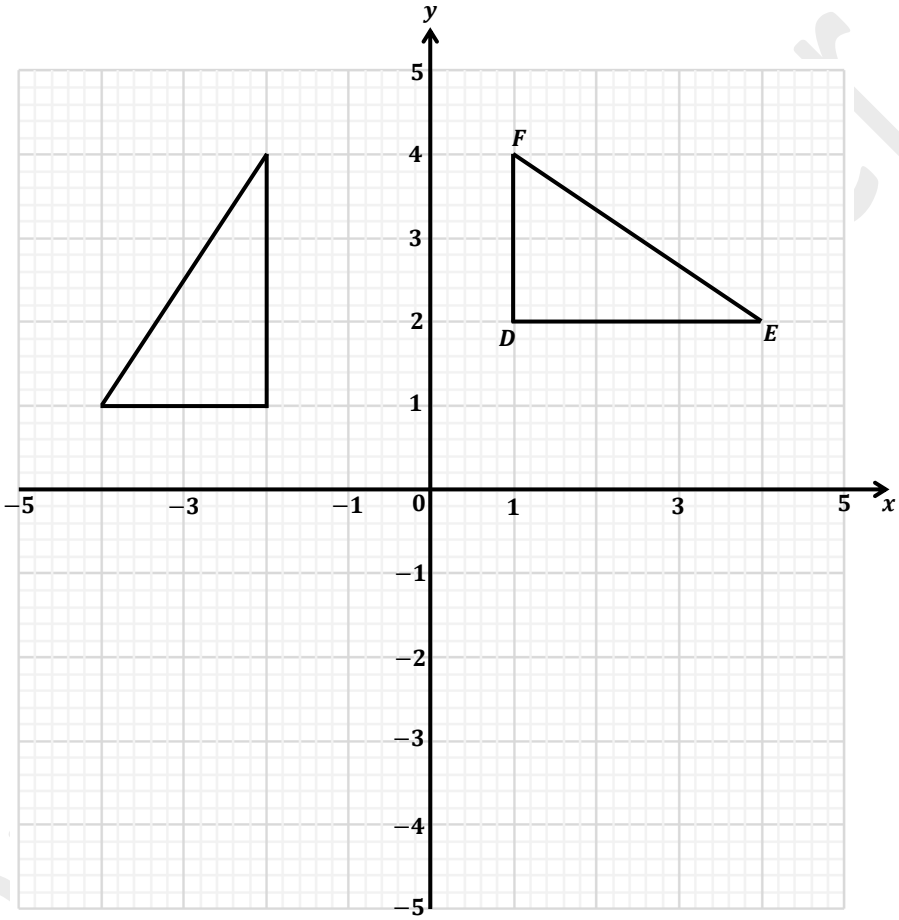
$\therefore$ The length of PQ is 19.2 cm.

(b) The graph below shows triangle DEF and its image $D'E'F'$ after a transformation.



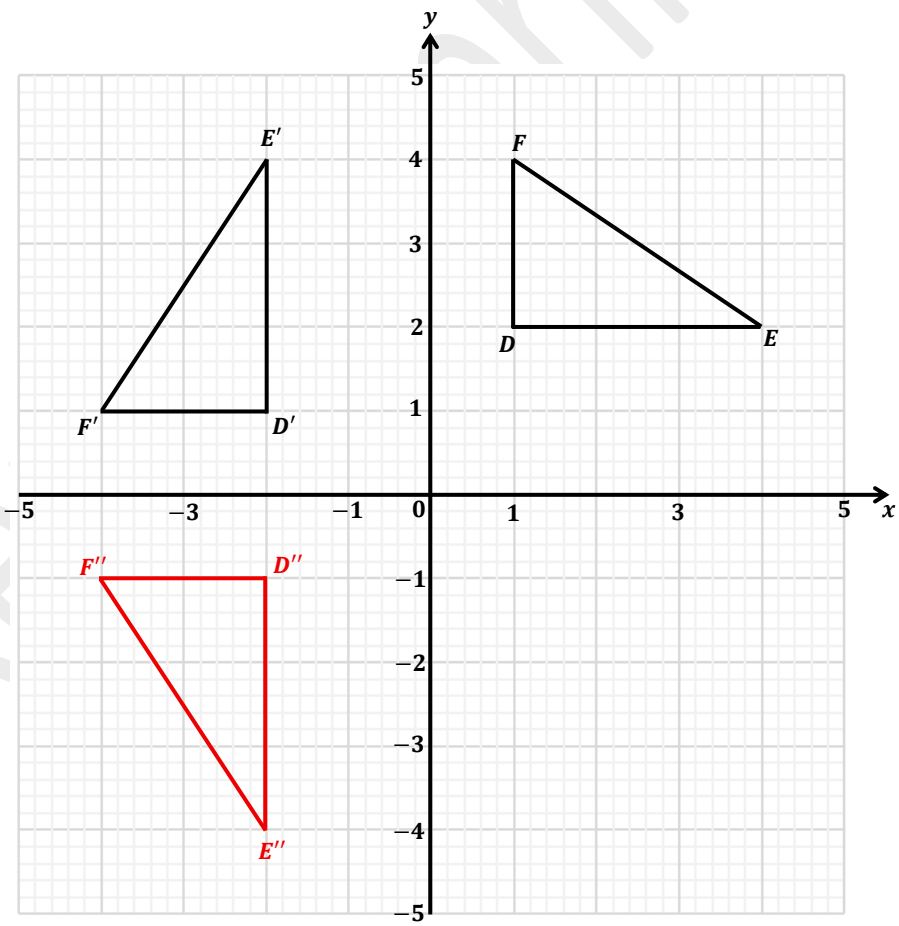
(i) State the coordinates of the point E. [1]

From the graph, the coordinates of the point E are $(4, 2)$.

(ii) Describe fully the transformation that maps triangle DEF to its image, $D'E'F'$. [3]

The transformation that maps triangle DEF to its image, $D'E'F$, is a 90° anti-clockwise rotation about the origin, $O(0, 0)$.

(iii) On the grid above, draw triangle $D"E"F"$, the reflection of triangle $D'E'F'$ in the x-axis. [2]



The coordinates of D''E''F'' are:

$D'' = (-2, -1)$

$E'' = (-2, -4)$

$F'' = (-4, -1)$

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6. (a) The scale on a map is 1: 25 000.

(i) Anderlin and Jersey are 31.8 cm apart on the map.

Determine, in km, the actual distance between Anderlin and Jersey. [2]

Anderlin and Jersey are 31.8 cm apart on the map.

The scale is 1 : 25 000.

1 cm on the map = 25 000 cm on the ground.

Actual distance = 31.8 × 25 000

= 795 000 cm

Now,

100 000 $cm = 1\ km$

$$1\ cm = \frac{1}{100\ 000}\ km$$

$$795\ 000\ cm = \frac{1}{100\ 000} \times 795\ 000$$

$$= 7.95\ km$$

∴ The actual distance between Anderlin and Jersey is 7.95 km.

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(ii) The actual distance between Clifton and James Town is 2.75 km.

How many units apart are they on the map? [2]

The actual distance between Clifton and James Town is 2.75 km.

Now,

$1\ km = 100\ 000\ cm$

$2.75\ km = 2.75 \times 100\ 000$

$= 275\ 000\ cm$

The scale is $1 : 25\ 000$.

$25\ 000\ cm$ on the ground represents $1\ cm$ on the map.

$275\ 000\ cm = \frac{275\ 000}{25\ 000}$

$= 11\ cm$

∴ Clifton and James Town is 11 units apart on the map.

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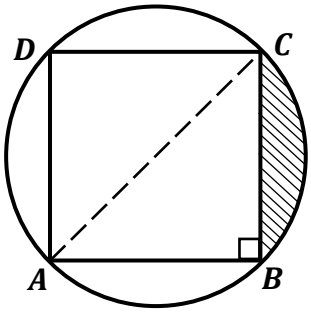
(b) The diagram below shows a square $ABCD$ drawn inside a circle. The vertices of the square lie on the circumference of the circle. The length of a side of the square is 11 cm.



(i) Show that the diameter of the circle is $11\sqrt{2}$ cm. [2]

Consider the triangle ABC. Using Pythagoras' Theorem,

$$(AC)^2 = (AB)^2 + (BC)^2$$

$$= (11)^2 + (11)^2$$

$$= 121 + 121$$

$$= 242$$

$$AC = \sqrt{242}$$

$$= \sqrt{121 \times 2}$$

$$= 11\sqrt{2}\ cm$$

$\therefore$ The diameter of the circle is $11\sqrt{2}\ cm$.

Q.E.D.

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Write your answers correct to 2 decimal places.

Calculate

(ii) the area of the circle [2]

Diameter of circle $= 11\sqrt{2}\ cm$

Radius of circle $= \frac{\text{Diameter}}{2}$

$= \frac{11\sqrt{2}}{2}\ cm$

Now,

Area of the circle $= \pi r^2$

$= \pi \left(\frac{11\sqrt{2}}{2}\right)^2$

$= \pi \times \frac{121 \times 2}{4}$

$= 190.09\ cm^2$ [to 2 decimal places]

(iii) the area of the square [1]

Area of the square $= s \times s$

$= 11 \times 11$

$= 121\ cm^2$

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(iv) the area of the shaded section [2]

There are four equal segments shown on the diagram.

Three of these are unshaded and 1 is shaded.

Area of these four equal segments = Area of circle − Area of square

$= 190.09 - 121$

$= 69.09\ cm^2$

Hence,

Area of the shaded region $= \frac{69.09}{4}$

$= 17.27\ cm^2$ [to 2 decimal places]

Total: 11 marks

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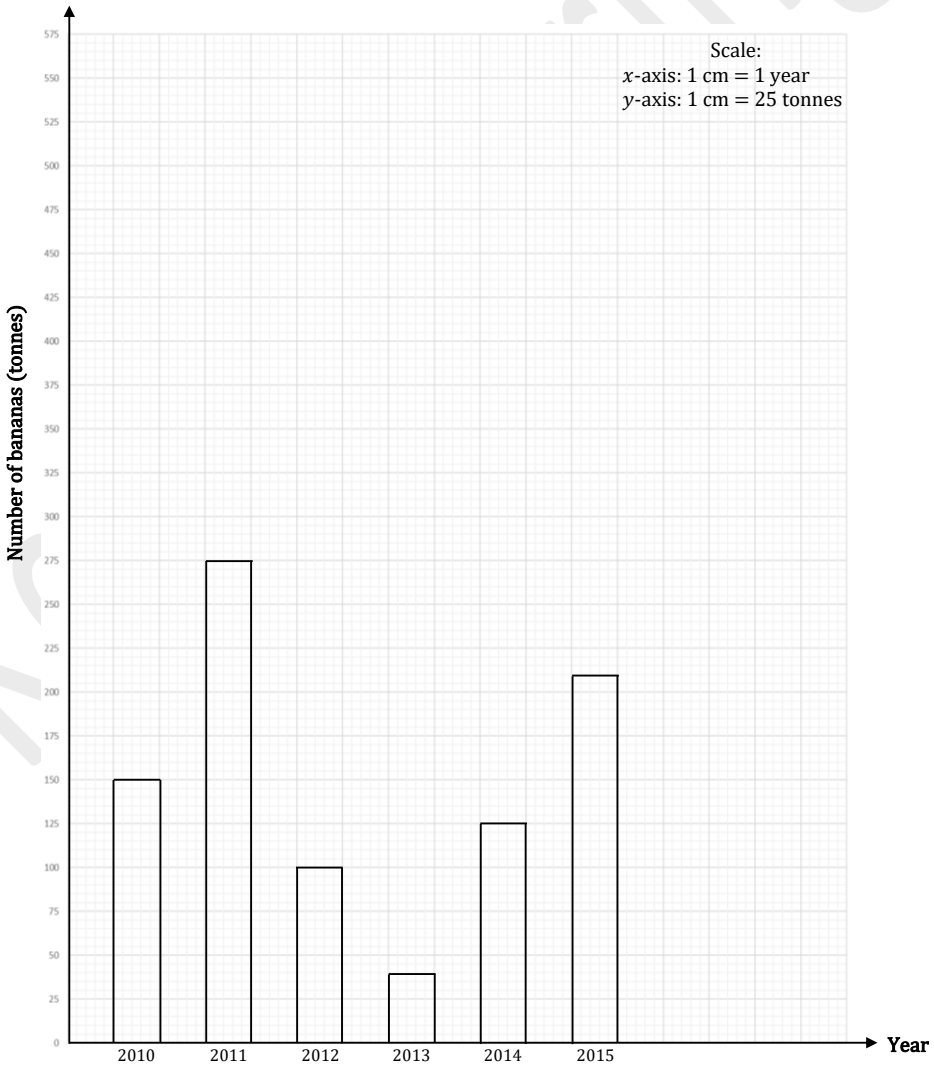
7. The table below shows the number of bananas, to the nearest tonne, produced annually on a farm over a period of 6 years.

Year	2010	2011	2012	2013	2014	2015
Production (tonnes)	150	275	100	40	125	210

(a) **On the graph paper provided on page 21**, draw a bar chart to represent the data given in the table above using a scale of 1 cm to represent 1 year on the x-axis and 1 cm to represent 25 tonnes on the y-axis. [4]



(b) Determine the range of the number of bananas produced between 2010 and 2015. [2]

Range = Highest value − Lowest value

= 275 − 40

= 235 tonnes

(c) (i) During which year was there the greatest production of bananas? [1]

From the bar chart, the year with the greatest production of bananas is 2011.

(ii) How is this information shown on the bar chart? [1]

The way the greatest production of bananas is shown on the bar chart is by the highest bar.

(d) (i) Between which two consecutive years was there the greatest **change** in the production of bananas? [1]

The greatest changed in the production of bananas occurred between the years 2011 and 2012.

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(ii) How is this information shown on the bar chart? [1]

This is shown on the bar chart with the difference between the respective heights of any two consecutive bars being the greatest.

(e) Give ONE reason why the bar chart is unsuitable for predicting the number of bananas produced in 2016. [1]

The bars on the graph do not show a consistent increasing or decreasing trend over the years. Therefore, it is not possible to make a reliable prediction for 2016.

Total: 11 marks

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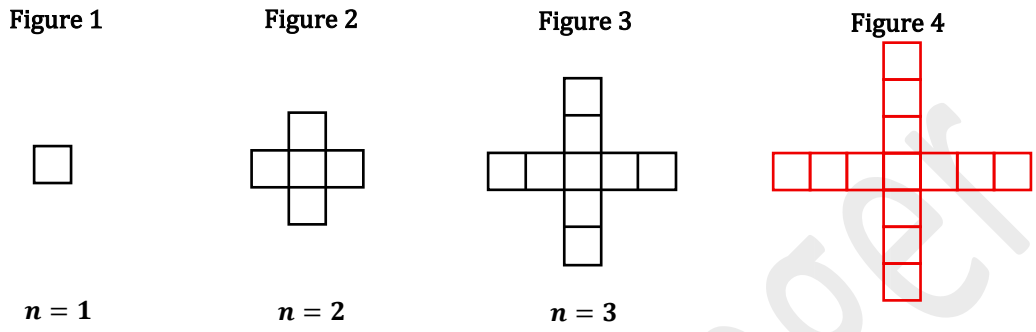
8. A sequence of figures is made up of unit squares with unit sides. The first three figures in the sequence are shown below.



(a) Draw Figure 4 of the sequence in the space provided above. [2]

Figure 4 has been drawn above.

(b) Study the pattern of numbers in each row of the table below. Each row relates to one of the figures in the sequence. Some rows have not been included in the table.

Complete the rows numbered (i), (ii), (iii) and (iv).

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	Figure	Number of Unit Squares	Perimeter of Figure	
	1	1	4	
	2	5	12	
	3	9	20	
(i)	4	13	28	[2]
(ii)	12	45	92	[2]
(iii)	30	117	236	[2]
(iv)	n	$4n - 3$	$8n - 4$	[2]

(iv) Consider the nth figure.

Number of unit squares $= 4n - 3$

Perimeter of figure $= 8n - 4$

(i) When $n = 4$,

Number of unit squares $= 4(4) - 3$

$= 16 - 3$

$= 13$

Perimeter of figure $= 8(4) - 4$

$= 32 - 4$

$= 28$

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(ii) The number of unit squares is 45.

So, we have,

$4n - 3 = 45$

$4n = 45 + 3$

$4n = 48$

$n = \frac{48}{4}$

$n = 12$

When $n = 12$,

Perimeter of figure $= 8(12) - 4$

$= 96 - 4$

$= 92$

(iii) When $n = 30$,

Number of unit squares $= 4(30) - 3$

$= 120 - 3$

$= 117$

Perimeter of figure $= 8(30) - 4$

$= 240 - 4$

$= 236$

Total: 10 marks

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SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) The table below shows pairs of values of x and y, where y is inversely proportional to x.

x	3	4	a	20
y	2	1.5	1.2	b

(i) Express y in terms of x and a constant k. [1]

y is inversely proportional to x.

$y \propto \frac{1}{x}$

$y = k \times \frac{1}{x}$

$y = \frac{k}{x}$ [where k is the constant of proportionality]

(ii) Calculate the value of the constant k. [1]

From the table, $y = 2$ when $x = 3$.

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Substituting $x = 3$ and $y = 2$ into $y = \frac{k}{x}$ gives:

$2 = \frac{k}{3}$

$k = 2 \times 3$

$k = 6$

$\therefore$ The value of the constant k is $k = 6$.

(iii) Determine the values of a and b. [2]

We have $y = \frac{6}{x}$.

When $y = 1.2$,

$1.2 = \frac{6}{x}$

$1.2x = 6$

$x = \frac{6}{1.2}$

$x = 5$

So, $a = 5$.

When $x = 20$,

$y = \frac{6}{20}$

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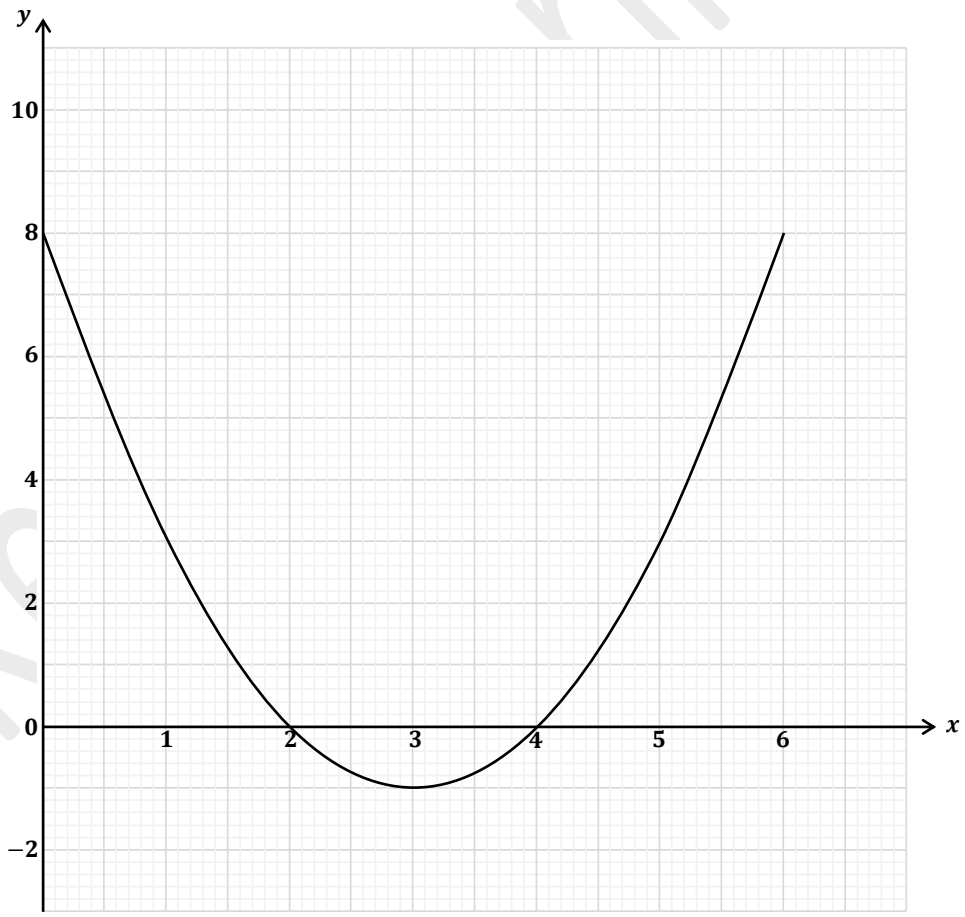
$y = 0.3$

So, $b = 0.3$.

$\therefore$ The values of a and b are $a = 5$ and $b = 0.3$.

(b) The diagram below shows the graph of the function $f(x) = x^2 - 6x + 8$ for values of x from 0 to 6.



(i) Use the graph to solve the equation $x^2 - 6x + 8 = 0$. [2]

From the graph, the curve cuts the x-axis at $x = 2$ and $x = 4$.

Hence, the solutions are $x = 2$ and $x = 4$.

(ii) Write down the coordinates of the minimum point in the form (x, y). [1]

From the graph, the minimum point is $(3, -1)$.

(iii) Write $x^2 - 6x + 8$ in the form $a(x + h)^2 + k$ where a, h and k are constants. [3]

$x^2 - 6x + 8$ which is in the form $ax^2 + bx + c$,

where $a = 1, \quad b = -6 \quad$ and $\quad c = 8$

$h = \frac{b}{2a}$

$h = \frac{-6}{2(1)}$

$h = \frac{-6}{2}$

$h = -3$

$k = \frac{4ac - b^2}{4a}$

$k = \frac{4(1)(8) - (-6)^2}{4(1)}$

$k = \frac{32 - 36}{4}$

$k = \frac{-4}{4}$

$k = -1$

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Alternatively,

$x^2 - 6x + 8$

$= (x^2 - 6x + 9) + 8 - 9$

$= (x - 3)^2 - 1$

$\therefore x^2 - 6x + 8 = (x - 3)^2 - 1$ which is in the form $a(x + h)^2 + k$,

where $a = 1, h = -3$ and $k = -1$.

(iv) On the same axes, draw the graph of the straight line

$g(x) = x - 2$. [3]

Let $y = x - 2$.

When $x = 0$,

$y = 0 - 2$

$y = -2$

When $x = 6$,

$y = 6 - 2$

$y = 4$

We plot the points $(0, -2)$ and $(6, 4)$ and connect them.

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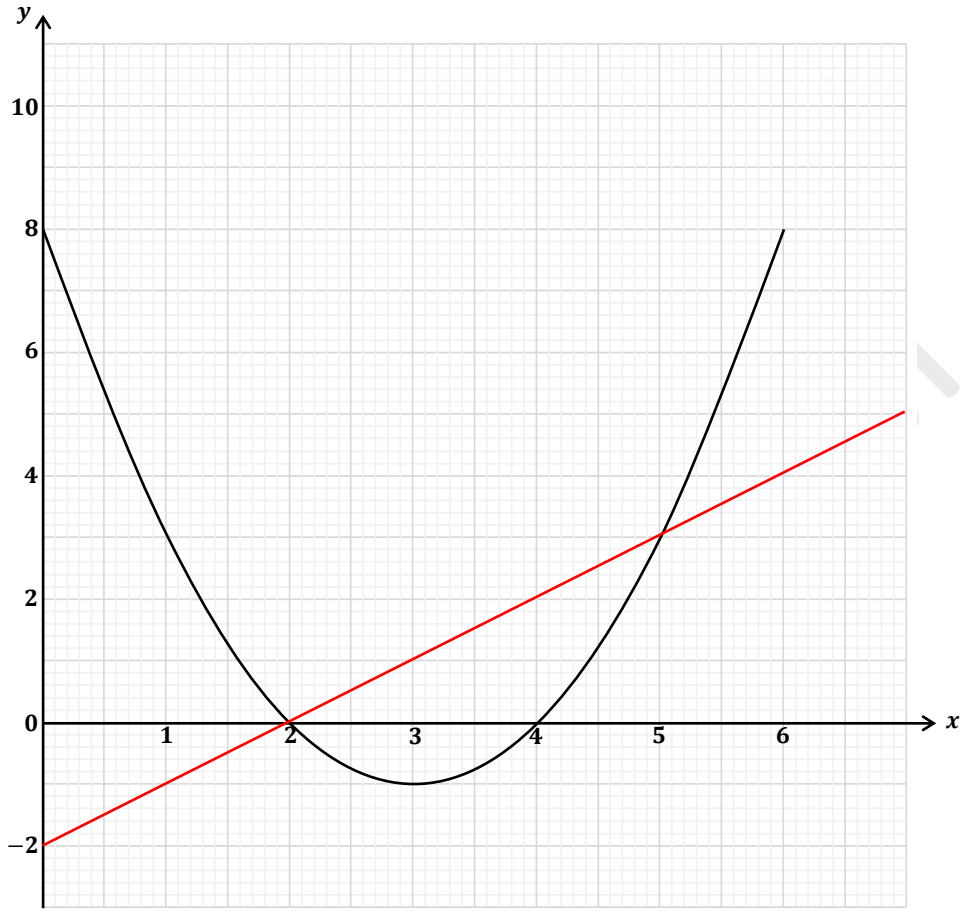


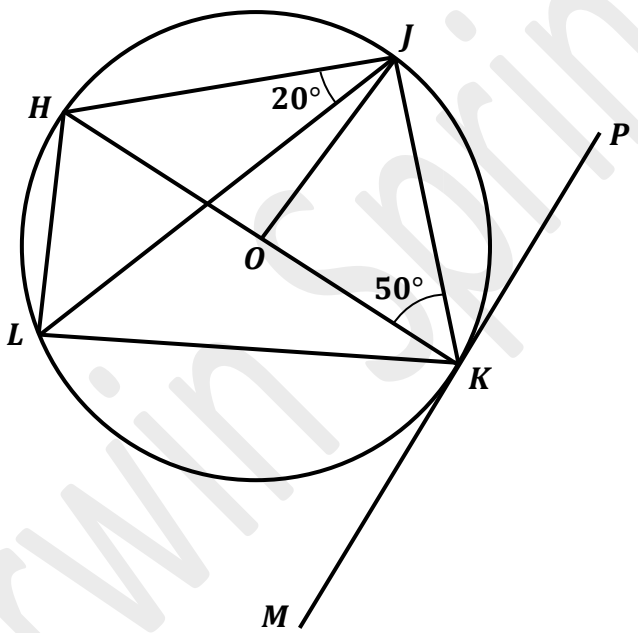
(v) Hence, solve the equation $x^2 - 6x + 8 = x - 2$. [2]

From the graph in (b)(iv) above, the graphs of $y = x^2 - 6x + 8$ and $y = x - 2$ intersect at $(2, 0)$ and $(5, 3)$.

$\therefore$ The solutions of $x^2 - 6x + 8 = x - 2$ are $x = 2$ and $x = 5$.

Total: 15 marks

10. (a) The diagram below, **not drawn to scale**, shows a circle with centre O. The vertices H, J, K and L of a quadrilateral lie on the circumference of the circle and PKM is a tangent to the circle at K. The measure of angle $H\hat{J}L = 20°$ and $J\hat{K}H = 50°$.



Calculate, **giving reasons for each step of your answer**, the measure of

(i) $H\hat{K}L$ [2]

Angles from the same chord, HL, in the same segment are equal.

$H\hat{K}L = H\hat{J}L$

$= 20°$

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(ii) $J\hat{O}K$ [2]

Consider triangle JOK.

Since OJ and OK are both radii of the same circle, then $OJ = OK$ and triangle JOK is an isosceles triangle.

The base angles in an isosceles triangle are equal.

$O\hat{J}K = O\hat{K}J$

$= 50°$

The angles in a triangle sum to 180°.

$J\hat{O}K = 180° - (50° + 50°)$

$= 180° - 100°$

$= 80°$

(iii) $J\hat{H}K$ [2]

The angle made by a tangent, MKP, to a circle and a chord, JK, at the point of contact is 90°. So, $O\hat{K}P = 90°$.

Now,

$J\hat{K}P = 90° - 50°$

$= 40°$

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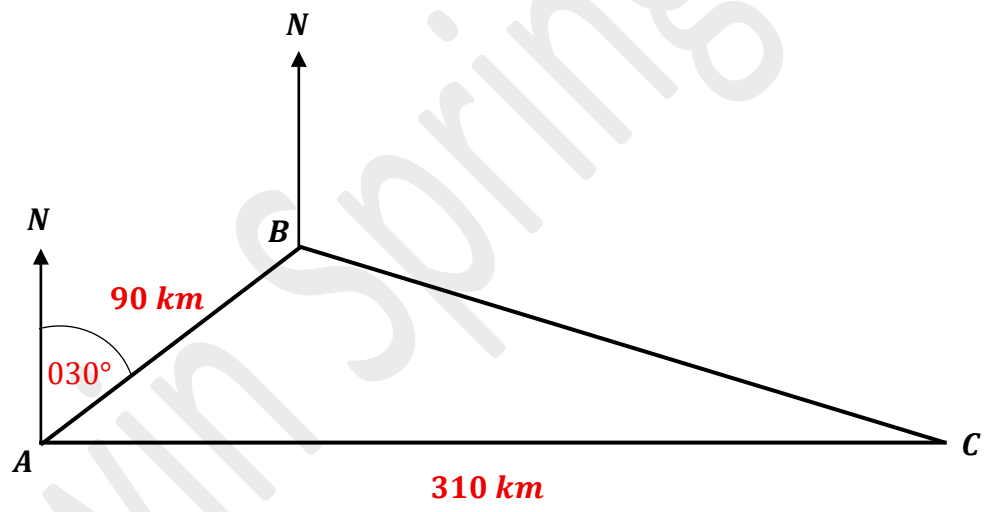
The angle between a tangent (MP) to a circle and a chord (JK) at the point of contact (K) is equal to the angle $(J\hat{H}K)$ in the alternate segment.

$\therefore J\hat{H}K = 40°$

(b) A ship travels from Akron (A) on a bearing of 030° to Bellville (B), 90 km away. It then travels to Comptin (C) which is 310 km due east of Akron (A), as shown in the diagram below.



(i) Indicate on the diagram the bearing of 030° and the distances 90 km and 310 km. [2]

Shown on the diagram above.

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(ii) Calculate, to the nearest km, the distance between Bellville (B) and Comptin (C). [2]

Using cosine rule,

$$(BC)^2 = (AB)^2 + (AC)^2 - 2(AB)(AC)\cos B\hat{A}C$$

$$= (90)^2 + (310)^2 - 2(90)(310)\cos 60°$$

$$= 76\,300$$

$$BC = \sqrt{76\,300}$$

$$= 276\ km \qquad \text{[to the nearest } km\text{]}$$

(iii) Calculate, to the nearest degree, the measure of $A\hat{B}C$. [2]

Using the cosine rule,

$$(AC)^2 = (AB)^2 + (BC)^2 - 2(AB)(BC)\cos A\hat{B}C$$

$$(310)^2 = (90)^2 + (276)^2 - 2(90)(276)\cos A\hat{B}C$$

$$96100 = 8100 + 76300 - 49680\cos A\hat{B}C$$

$$96100 = 84400 - 49680\cos A\hat{B}C$$

$$49680\cos A\hat{B}C = 84400 - 96100$$

$$49680\cos A\hat{B}C = -11700$$

$$\cos A\hat{B}C = \frac{-11700}{49680}$$

$$\cos A\hat{B}C = -0.236$$

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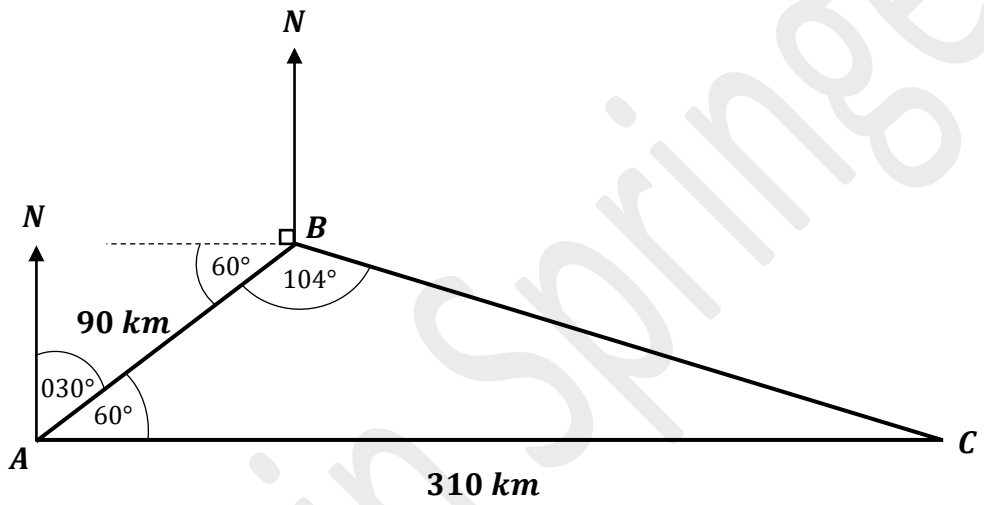
$A\hat{B}C = \cos^{-1}(-0.236)$

$A\hat{B}C = 104°$ (to the nearest degree)

(iv) Determine the bearing of Comptin (C) from Bellville (B). [3]



Bearing of C from $B = 360° - (104° + 60° + 90°)$

$= 360° - 254°$

$= 106°$ [to the nearest degree]

Total: 15 marks

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11. (a) The matrix $T = \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix}$ maps the point $P(2, 3)$ onto the point $Q(2, -3)$.

(i) Determine the values of c and d. [2]

The matrix $T = \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix}$ maps the point $P(2, 3)$ onto the point $Q(2, -3)$.

So, we have,

$$\begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} (c \times 2) + (0 \times 3) \\ (0 \times 2) + (d \times 3) \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} 2c + 0 \\ 0 + 3d \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} 2c \\ 3d \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

Equating corresponding entries gives:

$2c = 2$	or	$3d = -3$
$c = \frac{2}{2}$		$d = \frac{-3}{3}$
$c = 1$		$d = -1$

$\therefore$ The values of c and d are $c = 1$ and $d = -1$.

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(ii) Determine the image of $(-5, 4)$ under the transformation T. [1]

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} -5 \\ 4 \end{pmatrix} = \begin{pmatrix} (1 \times -5) + (0 \times 4) \\ (0 \times -5) + (-1 \times 4) \end{pmatrix}$$

$$= \begin{pmatrix} -5 \\ -4 \end{pmatrix}$$

$\therefore$ The image is $(-5, -4)$.

(iii) Describe fully the transformation T. [2]

$T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ describes a reflection in the x-axis.

(iv) Find the matrix that maps the point Q back onto the point P. [2]

The matrix that maps the point Q back onto the point P is the inverse of T, T^{-1}.

$$T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\det(T) = ad - bc$$

$$= (1)(-1) - (0)(0)$$

$$= -1 - 0$$

$$= -1$$

$$adj(T) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

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Now,

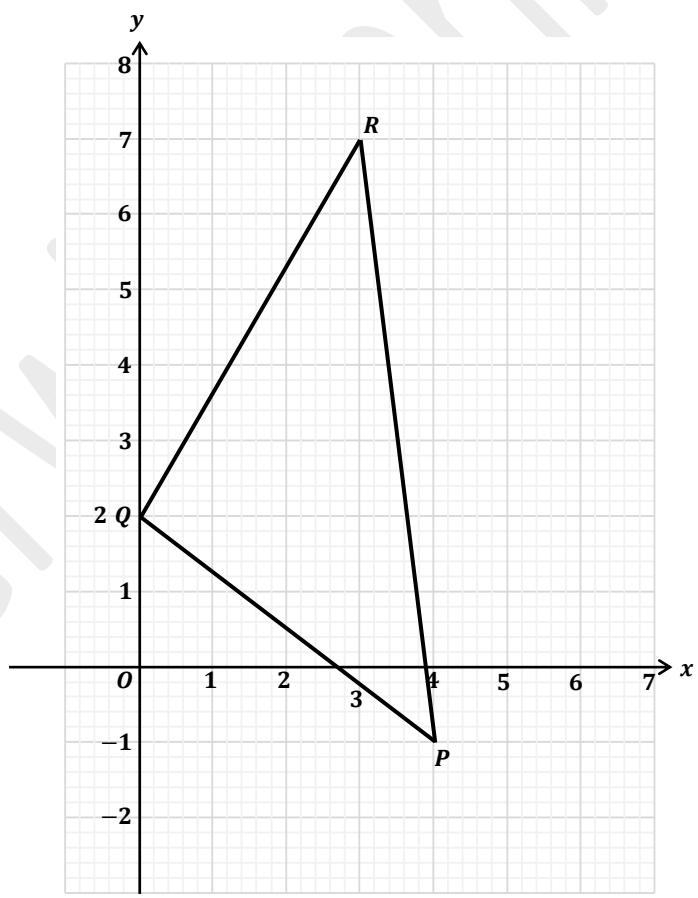
$$T^{-1} = \frac{1}{\det(T)} \times adj(T)$$

$$= \frac{1}{-1}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= -1\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

(b) The graph below shows three points P, Q and R, relative to the origin, O.



(i) Write as a column vector in the form $\begin{pmatrix} x \\ y \end{pmatrix}$

- the vector $\overrightarrow{OP}$ [1]
- the vector $\overrightarrow{QR}$ [2]

The point P has coordinates $P(4, -1)$.

Hence, $\overrightarrow{OP} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$ which is of the form $\begin{pmatrix} x \\ y \end{pmatrix}$ where $x = 4$ and $y = -1$.

Now,

The point Q has coordinates $Q(0, 2)$.

So, $\overrightarrow{OQ} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$.

The point R has coordinates $R(3, 7)$.

So, $\overrightarrow{OR} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$.

Hence,

$$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ}$$

$$= \begin{pmatrix} 3 \\ 7 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 - 0 \\ 7 - 2 \end{pmatrix}$$

$= \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ which is of the form $\begin{pmatrix} x \\ y \end{pmatrix}$ where $x = 3$ and $y = 5$.

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(ii) Determine the magnitude of the vector $\overrightarrow{QR}$. [2]

$\overrightarrow{QR} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$.

$|\overrightarrow{QR}| = \sqrt{(3)^2 + (5)^2}$

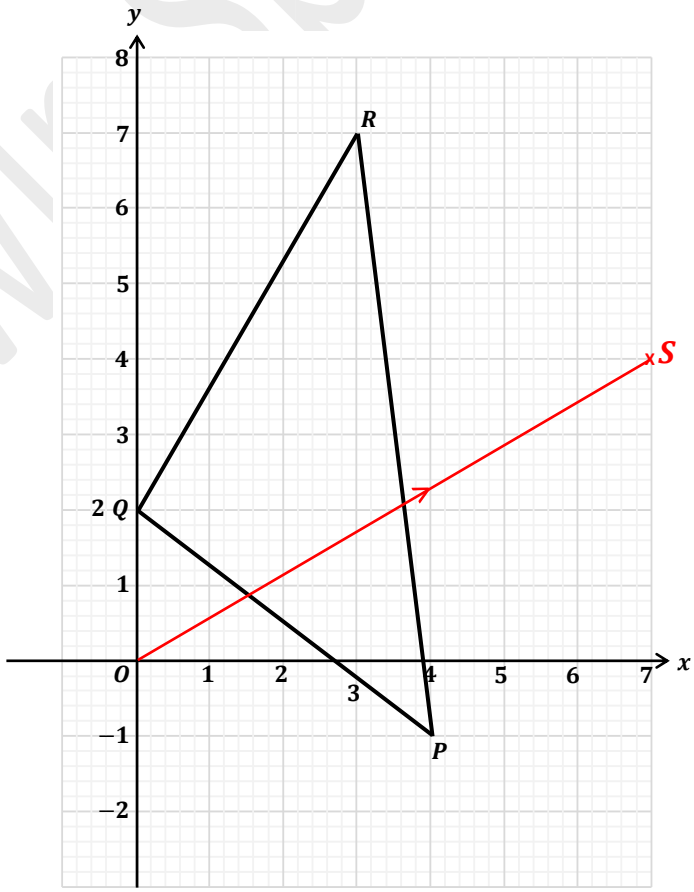
$= \sqrt{9 + 25}$

$= \sqrt{34}$ units

(iii) On the graph provided on page 33, draw the vector $\overrightarrow{OS} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$. Show that $PQRS$ is a parallelogram. [3]

$\overrightarrow{OS} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$

$\therefore S = (7, 4)$.



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Using the vector triangle law,

$\overrightarrow{QP} = \overrightarrow{OP} - \overrightarrow{OQ}$

$= \begin{pmatrix} 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} 4 - 0 \\ -1 - 2 \end{pmatrix}$

$= \begin{pmatrix} 4 \\ -3 \end{pmatrix}$

Using the vector triangle law,

$\overrightarrow{RS} = \overrightarrow{OS} - \overrightarrow{OR}$

$= \begin{pmatrix} 7 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

$= \begin{pmatrix} 7 - 3 \\ 4 - 7 \end{pmatrix}$

$= \begin{pmatrix} 4 \\ -3 \end{pmatrix}$

Therefore, $\overrightarrow{QP} = \overrightarrow{RS}$, that is, $|\overrightarrow{QP}| = |\overrightarrow{RS}|$ and $\overrightarrow{QP}$ is parallel to $\overrightarrow{RS}$.

If the opposite sides of a quadrilateral are parallel and equal, then the quadrilateral is a parallelogram.

Hence, $PQRS$ is a parallelogram.

Q.E.D.

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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